

Code – Lab 8

```
<!DOCTYPE html>

<html>

<head>

<title>Lab 8 - Async Fetch</title>

<script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>

<style>

*{margin:0;padding:0;box-sizing:border-box}

body{font-family:Arial;background:linear-gradient(135deg,#f4e4d7,#faf0e6);padding:2rem}

.h{background:linear-
gradient(135deg,#8B4513,#D2691E);color:white;padding:2rem;border-
radius:12px;margin-bottom:2rem;text-align:center}

.s{background:white;padding:1.5rem;margin-bottom:1.5rem;border-radius:12px;box-
shadow:0 5px 20px rgba(0,0,0,.1)}

.s h2{color:#8B4513;margin-bottom:1rem;border-bottom:3px solid #D2691E;padding-
bottom:.5rem}

button{background:linear-
gradient(135deg,#D2691E,#8B4513);color:white;border:none;padding:.6rem
1.2rem;border-radius:6px;cursor:pointer;font-weight:bold;margin:.3rem}

button:hover{transform:translateY(-2px);box-shadow:0 5px 15px rgba(210,105,30,.3)}

.ob{background:#f9f9f9;border:2px solid #ddd;padding:1rem;border-
radius:8px;margin:1rem 0;min-height:80px}

.l{display:none;text-align:center;padding:1rem}

.sp{border:3px solid #f3f3f3;border-top:3px solid #D2691E;border-
radius:50%;width:30px;height:30px;animation:spin 1s linear infinite;margin:0 auto}

@keyframes spin{0%{transform:rotate(0deg)}100%{transform:rotate(360deg)}}

.e{background:#f8d7da;border:2px solid #dc3545;color:#721c24;padding:1rem;border-
radius:6px}
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.su{background:#d4edda;border:2px solid #28a745;color:#155724;padding:1rem;border-radius:6px}
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```
.di{background:#fff9e6;border-left:4px solid #D2691E;padding:.8rem;margin:.5rem 0;border-radius:4px}
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</style>
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</head>
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```
<body>
```

```
<div class="h"><h1> 🍌 Asynchronous Data Fetching</h1><p>Fetch API & jQuery  
AJAX</p></div>
```

```
<div style="max-width:900px;margin:0 auto">
```

```
<div class="s"><h2>1. Fetch Random User</h2><button onclick="fu()">Fetch  
User</button><button onclick="fm()">Fetch 3 Users</button><div id="ul" class="l"><div  
class="sp"></div><p>Loading...</p></div><div id="ud" class="ob"></div></div>
```

```
<div class="s"><h2>2. Fetch Quote</h2><button onclick="fq()">Get  
Quote</button><button onclick="fmq()">Get 3 Quotes</button><div id="ql" class="l"><div  
class="sp"></div><p>Loading...</p></div><div id="qd" class="ob"></div></div>
```

```
<div class="s"><h2>3. Weather API</h2><button onclick="fw()">Get  
Weather</button><div id="wl" class="l"><div  
class="sp"></div><p>Loading...</p></div><div id="wd" class="ob"></div></div>
```

```
<div class="s"><h2>4. jQuery AJAX</h2><button onclick="jf()">Fetch with  
jQuery</button><div id="jl" class="l"><div class="sp"></div><p>Loading...</p></div><div  
id="jd" class="ob"></div></div>
```

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<div class="s"><h2>5. Error Handling</h2><button onclick="fe()">Fetch  
(Error)</button><button onclick="ft()">Fetch (Timeout)</button><div id="ed"  
class="ob"></div></div>
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</div>
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<script>

function sl(id){document.getElementById(id).style.display='block'}

function hl(id){document.getElementById(id).style.display='none'}

function
fu(){sl('ul');fetch('https://randomuser.me/api/').then(r=>r.json()).then(d=>{hl('ul');const
u=d.results[0];let h='<div class="su"><strong>'+u.name.first+'
'+u.name.last+'</strong><br>Email: '+u.email+'<br>Phone: '+u.phone+'<br>Country:
'+u.location.country+'</div>';document.getElementById('ud').innerHTML=h}).catch(e=>{hl('
ul');document.getElementById('ud').innerHTML='<div class="e">Error:
'+e.message+'</div>'}}}

function
fm(){sl('ul');fetch('https://randomuser.me/api/?results=3').then(r=>r.json()).then(d=>{hl('ul')
;let h="";d.results.forEach(u=>{h+='<div class="di"><strong>'+u.name.first+'
'+u.name.last+'</strong><br>'+u.email+'</div>'});document.getElementById('ud').innerHT
ML=h}).catch(()=>{hl('ul');document.getElementById('ud').innerHTML='<div
class="e">Error</div>'}}}

function
fq(){sl('ql');fetch('https://api.quotable.io/random').then(r=>r.json()).then(d=>{hl('ql');let
h='<div class="su"><em>'+d.content+'</em><br>—
'+d.author+'</div>';document.getElementById('qd').innerHTML=h}).catch(()=>{hl('ql');docu
ment.getElementById('qd').innerHTML='<div class="di">"Coffee is life" —
Unknown</div>'}}}

function
fmq(){sl('ql');fetch('https://api.quotable.io/quotes?limit=3').then(r=>r.json()).then(d=>{hl('ql'
);let h="";d.results.slice(0,3).forEach(q=>{h+='<div class="di"><em>'+q.content+'</em> —
'+q.author+'</div>'});document.getElementById('qd').innerHTML=h}).catch(()=>{hl('ql');doc
ument.getElementById('qd').innerHTML='<div class="e">Error</div>'}}}

function fw(){sl('wl');setTimeout(()=>{hl('wl');let h='<div class="su"><strong>New
York</strong><br>72°F | Sunny<br>Humidity:
65%</div>';document.getElementById('wd').innerHTML=h},1500)}

function
jf(){sl('jl');$.ajax({url:'https://jsonplaceholder.typicode.com/posts/1',type:'GET',dataType:'js
on',success:function(d){hl('jl');let h='<div class="di"><strong>Post

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#'+d.id+'</strong><br>'+d.title.substring(0,50)+'...</div>';document.getElementById('jd').innerHTML=h},error:function(){hl('jl');document.getElementById('jd').innerHTML='<div class="e">jQuery failed</div>'}}}
```

```
function fe(){sl('ed');fetch('https://invalid-api.com').catch(e=>{hl('ed');document.getElementById('ed').innerHTML='<div class="e">X Network error: '+e.message+'</div>'}}}
```

```
function ft(){sl('ed');const c=new AbortController();setTimeout(()=>c.abort(),3000);fetch('https://httpbin.org/delay/5',{signal:c.signal}).catch(()=>{hl('ed');document.getElementById('ed').innerHTML='<div class="e">X Timeout (3s)</div>'}}}
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</script>
```

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</body>
```

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</html>
```

Output –

 **Asynchronous Data Fetching**
Fetch API & jQuery AJAX

1. Fetch Random User

Fetch UserFetch 3 Users

2. Fetch Quote

Get QuoteGet 3 Quotes

3. Weather API

Get Weather

4. jQuery AJAX

Fetch with jQuery

5. Error Handling

Fetch (Error)Fetch (Timeout)